A local obstruction to a quartic pole equation

Abstract

The equation $y^2 = T^2 + \kappa R(1 + Rs^2)^2$ gives the strict valuation inequality required to extract local tests from a reciprocal trace. Nevertheless, with $\kappa = 1$ its associated genus-one equation has no point over $\mathbb{Q}_{1013}$ for an explicit infinite family of prepared positive inputs tending to zero. The inputs have square reduced denominators and satisfy all prescribed squareclass and valuation conditions at $2, 3, 17$. Thus these preparation conditions alone do not give completeness for this equation. More generally, no finite set of fixed nonzero rational coefficients κ is complete on all prepared inputs. The result does not decide completeness on a subfamily selected together with the comparison primes.

1 The valuation implication

We normalize the valuation v_p of $\mathbb{Q}_p$ by $v_p(p) = 1$ and put $v_p(0) = +\infty$.

Lemma 1.1. *Let p be an odd prime and let $R, \kappa \in \mathbb{Q}_p^\times$ satisfy $v_p(\kappa) = 0$ and $\nu = v_p(R) \in 2\mathbb{Z} + 1$. If $y, s, T \in \mathbb{Q}_p$ satisfy*

$$y^2 = T^2 + \kappa R(1 + Rs^2)^2, \tag{1}$$

then $T \neq 0$ and

$$2v_p(T) < \nu. \tag{2}$$

Proof. If $s \neq 0$, then $v_p(Rs^2) = \nu + 2v_p(s)$ is odd and hence nonzero. Therefore

$$u := v_p(1 + Rs^2) = \min(0, \nu + 2v_p(s)) \leq 0.$$

For $s = 0$ the same conclusion holds with $u = 0$. In particular $1 + Rs^2 \neq 0$. The second term on the right of (1) has the odd valuation $\nu + 2u$. If $T = 0$, this cannot be the valuation of a square. If $T \neq 0$, its square has even valuation and cannot cancel the other term. The smaller valuation must be even, so $2v_p(T) < \nu + 2u \leq \nu$. □

Corollary 1.2. *Let $K/\mathbb{Q}$ be quadratic and unramified at p, and let $W \in K^\times$. Put*

$$T = \mathrm{Tr}_{K/\mathbb{Q}}(W^{-1}), \qquad \lambda_p = \max_{\mathfrak{p}|p} v_{\mathfrak{p}}(W),$$

where the local valuations have $v_{\mathfrak{p}}(p) = 1$. Under the hypotheses of Lemma 1.1, a rational solution of (1) implies

$$\lambda_p > -\nu/2.$$

Proof. At a split prime the two terms of the trace have valuations at least $-\lambda_p$. At an inert prime conjugation preserves the unique valuation and gives the same conclusion. The nonarchimedean triangle inequality yields $v_p(T) \geq -\lambda_p$. Apply (2). □

Since $T \neq 0$, the changes of variables

$$b = R/T^2, \qquad v = Ts, \qquad Y = y/T$$

transform (1) exactly into

$$Y^2 = 1 + \kappa b(1 + bv^2)^2. \tag{3}$$

For $b, \kappa > 0$ the polynomial on the right is a separable quartic: its leading and constant coefficients are nonzero, and the discriminant of its quadratic expression in v^2 is $-4\kappa b^3 \neq 0$. Its smooth projective double cover of the v-line consequently has genus one.

2 A local obstruction with even negative valuation

Lemma 2.1. *Let $p \equiv 1 \pmod 4$ be prime. Suppose that κ is a p-adic unit with square residue, and write*

$$b = p^{-2m}u, \qquad m \geq 1, \quad u \in \mathbb{Z}_p^{\times},$$

with u having nonsquare residue. Then (3) has no solution in $\mathbb{Q}_p$. Its smooth projective model also has no $\mathbb{Q}_p$-point.

Proof. Since $v_p(\kappa b) < 0$, one has $1 + \kappa b \neq 0$. The discriminant $-4\kappa b^3$ is nonzero, so the quartic is separable even when κ is negative. Consider $v \neq 0$. If $v_p(v) < m$, then $v_p(bv^2) < 0$ and $v_p(1 + bv^2) = v_p(bv^2) < 0$. If $v_p(v) > m$, then $v_p(1 + bv^2) = 0$. In the remaining case $v_p(v) = m$, the residue of bv^2 is a nonsquare. Since -1 is a square modulo p, the sum $1 + bv^2$ is a unit. The case $v = 0$ also gives a unit. Thus in every case

$$v_p(1 + bv^2) \leq 0, \qquad 1 + bv^2 \neq 0.$$

It follows that $\kappa b(1 + bv^2)^2$ has negative even valuation and nonsquare leading unit. Adding one preserves both properties. The result cannot be a square in $\mathbb{Q}_p$.

At infinity, the two points of the smooth quartic are defined over the extension obtained by adjoining a square root of the leading coefficient κb^3. Its valuation is even and its leading unit is again a nonsquare. Neither point is defined over $\mathbb{Q}_p$. □

3 Prepared counterexamples

Definition 3.1. A rational number b is prepared if its reduced denominator is a square and

$$0 < b < 17, \qquad b \in \mathbb{Q}_2^{\times 2} \cap \mathbb{Q}_3^{\times 2} \cap \mathbb{Q}_{17}^{\times 2},$$
$$v_2(b) \geq 3, \qquad v_3(b) \geq 1, \qquad v_{17}(b) \geq 2.$$

These are the preparation conditions of the pole-realization input in [1, Section 3].

Theorem 3.2. *For every integer $m \geq 1$, the number*

$$b_m = 409 \left(\frac{204}{1013^m} \right)^2$$

is prepared, but the smooth projective curve

$$Y^2 = 1 + b_m(1 + b_m v^2)^2 \tag{4}$$

has no point over $\mathbb{Q}_{1013}$ and hence no rational point. The numbers b_m are distinct and tend to zero.

Proof. Trial division certifies that 1013 is prime, and $\gcd(409 \cdot 204, 1013) = 1$. Thus the reduced denominator of b_m is 1013^{2m}. At $m = 1$ one has

$$b_1 = \frac{17020944}{1026169}, \qquad 17 \cdot 1026169 - 17020944 = 423929 > 0.$$

Every b_m lies in $(0, 17)$, and

$$v_2(b_m) = 4, \qquad v_3(b_m) = 2, \qquad v_{17}(b_m) = 2.$$

The congruences $409 \equiv 1 \pmod 8$, $409 \equiv 1 \pmod 3$ and $409 \equiv 1 \pmod{17}$ show that 409 is a square in each of $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$. Its product with the displayed rational square therefore satisfies all three local squareclass conditions.

At $p = 1013$ one has $p \equiv 1 \pmod 4$ and

$$409^{506} \equiv 1012 \pmod{1013}.$$

The unit $409 \cdot 204^2$ is consequently a nonsquare modulo 1013, and $v_{1013}(b_m) = -2m$. Lemma 2.1, with $\kappa = 1$, applies. Finally $b_{m+1} = b_m/1013^2$, proving the last assertion. □

Corollary 3.3. *Let C be a finite nonempty subset of $\mathbb{Q}^\times$. There are arbitrarily small prepared b for which none of the equations* (3)*, with $\kappa \in C$, has a rational point, even on its smooth projective model.*

Proof. Write $\kappa = a_\kappa/d_\kappa$ in reduced form with $d_\kappa > 0$, and put

$$M = 408 \prod_{\kappa \in C} |a_\kappa d_\kappa|.$$

Choose a prime divisor p of the cyclotomic value $\Phi_M(M) > 1$. It satisfies $p \nmid M$, since $\Phi_M(0) = 1$. The cyclotomic order lemma gives $p \equiv 1 \pmod M$: in characteristic p the polynomial $x^M - 1$ is separable, so its cyclotomic factors are pairwise coprime and a root of Φ_M has exact order M. In particular $p \equiv 1 \pmod 8$ and $p \equiv 1 \pmod r$ for every odd prime divisor r of a numerator or denominator above. Quadratic reciprocity shows that each such r, as well as -1 and 2, is a square modulo p. Thus every $\kappa \in C$ is a p-adic unit with square residue.

The Chinese remainder theorem supplies a positive integer c which is congruent to one modulo 408 and to a nonzero nonsquare modulo p. For all sufficiently large m, the numbers

$$b_m = c(204/p^m)^2$$

are in $(0, 17)$, have reduced denominator p^{2m} and valuations $4, 2, 2$ at $2, 3, 17$, and are squares in the three corresponding completions. Hence they are prepared. At p they have valuation $-2m$ and nonsquare leading unit $c \cdot 204^2$. Lemma 2.1 applies simultaneously to every $\kappa \in C$, and $b_m \to 0$. □

The valuation implication in Corollary 1.2 is valid independently of this failure of completeness. Theorem 3.2 shows that the stated preparation conditions do not suffice to replace the pole-realization theorem by (3) with $\kappa = 1$; Corollary 3.3 excludes any finite choice of fixed nonzero rational coefficients. Neither result shows that these counterexamples occur among the inputs obtained from simultaneous comparison-prime selection. No variable-count reduction follows without a realization argument on the required selected inputs.

4 Exact verification

The standard-library Python program

`check_quartic_pole_obstruction.py`

certifies primality, the Euler congruence, the preparation congruences, the exact first input, and the valuation case distinctions. It also checks every residue modulo 1013 in the potentially cancelling case and saves its output in

`quartic-pole-obstruction.json`.

The proofs above establish the assertions for every p-adic argument and every $m \geq 1$; the certificate records the finite arithmetic used in those proofs. No rank computation or conjectural input is needed for the obstruction.

References

[1] Project repository, *Arithmetic dependencies of the defining-polynomial constructions*, `arithmetic_dependencies_20261108_v8.tex`, Section 3.